

Propositional Equivalences

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Key Propositional Equivalences

Identity Laws

$$p \wedge \text{true} \equiv p$$

$$p \vee \text{false} \equiv p$$

Domination Laws

$$p \vee \text{true} \equiv \text{true}$$

$$p \wedge \text{false} \equiv \text{false}$$

Double Negation Law

$$\neg(\neg p) \equiv p$$

Commutative Laws

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

Idempotent Laws

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Absorption Laws

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

Negation Laws

$$p \vee \neg p \equiv \text{true}$$

$$p \wedge \neg p \equiv \text{false}$$

See Rules of Inference for more details.